

RESEARCH PAPER · DOUBLE-BLIND SUBMISSION

PharmaGA.

A smarter algorithm that finds the best drug combination for each individual patient.



01 • PROBLEM

Picking the right drug is hard — and getting harder.

There are over 10,000 approved drugs. Doctors often prescribe 2–4 at once. That means billions of possible combinations — far too many to test one by one.

Two trends are making this harder to solve.

FORCE ONE

Pharmacogenomics

A patient's genes can make a drug work brilliantly — or be **dangerous**. The same drug, very different outcomes.

FORCE TWO

Drug repurposing

We already have thousands of approved drugs. The challenge is **choosing the right one** for this patient, now.

02 • RESEARCH GAP

Current approaches have three big gaps.

01

Single-objective formulations

They combine all goals into one score — hiding important trade-offs from the doctor.

02

Blind genetic operators

They mix drugs randomly, breaking combinations that work well together for biological reasons.

03

Expensive oracle calls

Simulating how drugs behave is slow and costly — limiting how many options can be tested.

THE GAP

No existing tool picks drugs for a *specific patient*, weighs multiple goals, and runs at reasonable speed.

03 · CONTRIBUTIONS

Five contributions at a glance.

- c1** Patient-conditional MOO
Optimizes four clinical goals at once, tailored to each patient.
- c2** Mechanism-Complementarity Crossover
A new crossover method that keeps drug combinations that make biological sense.
- c3** Two-stage surrogate evaluation
Gradient-boosted approximator with uncertainty-driven oracle re-evaluation.
- c4** Personalization fidelity metric
A new score that checks whether results are truly tailored to the patient — not just generally good.
- c5** Open benchmark
1,842 FDA-approved & investigational compounds · 500 simulated patient profiles · 4 therapeutic areas.

04 · FORMULATION

How we define the problem.

ENCODING

$$s = \langle (d_{i_1}, \delta_{i_1}), \dots, (d_{i_k}, \delta_{i_k}) \rangle, k \leq K_{\max}$$

Each regimen is a list of up to 4 (drug, dose) pairs.

OBJECTIVE

$$\min_{s \in S_p} F(s, p) = \langle -f_1, f_2, f_3, f_4 \rangle(s, p)$$

Rules: no dangerous drugs for this patient · no duplicate mechanisms · max 4 drugs.

OUTPUT

$$P \star_p = \{s \in S_p : \nexists s', F(s', p) < F(s, p)\}$$

A set of best options — each one optimal in a different way. The doctor chooses.

PATIENT PROFILE $X_p \in \mathbb{R}^Q$

Everything we know about the patient.

- Demographic age, sex, BMI, eGFR
- Biochemical labs, comorbidity vector
- Pharmacogenomic CYP2D6 · CYP2C19
- TPMT · DPYD · HLA-B
- Clearance renal · hepatic indices

The goals *and* the constraints depend on who the patient is. That's the personalization.

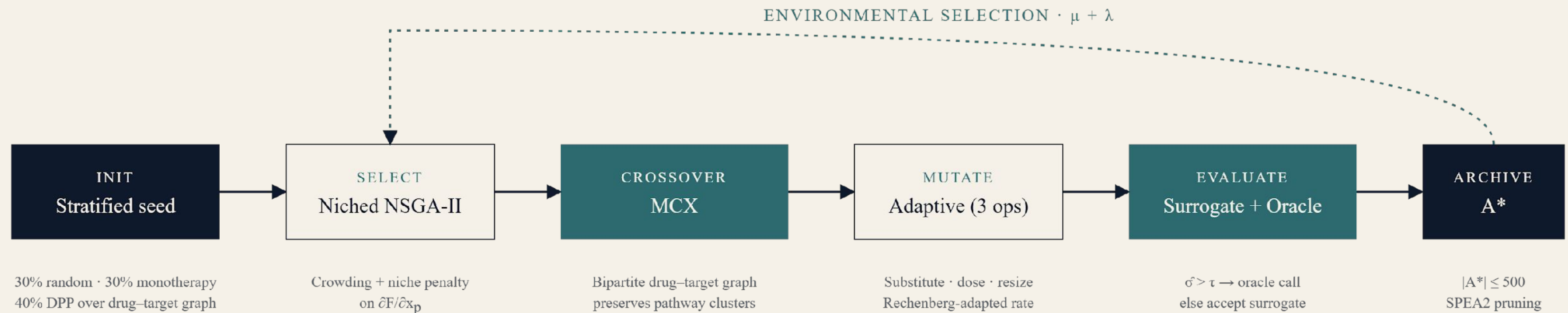
05 · OBJECTIVES

Four goals — balanced, not blended.

f_1 Predicted efficacy How likely is this drug combination to work for this patient? MAXIMIZE ↑	f_2 Toxicity burden How much harm could it do? Heart, liver, kidney, and rare-reaction risks combined. MINIMIZE ↓	f_3 PK mismatch Does the patient's body process this drug poorly due to their genes? MINIMIZE ↓	f_4 DDI risk Do the drugs in the combination interfere with each other? MINIMIZE ↓
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06 · ALGORITHM

How the algorithm works, step by step.



Crossover that respects pharmacology.

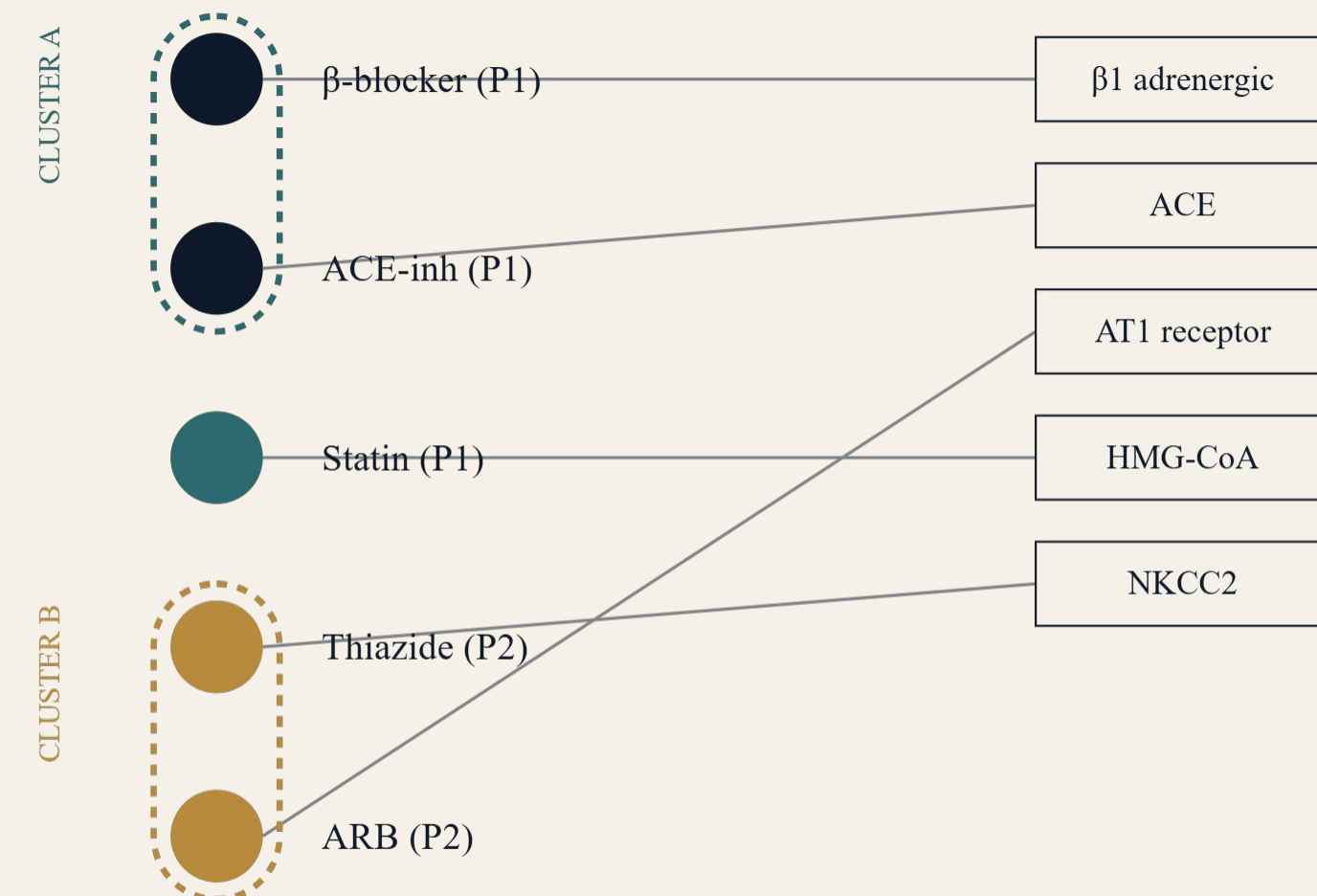
Standard mixing splits drug pairs that belong together — like separating a β -blocker and ACE inhibitor that work as a team. MCX keeps those pairs intact.

PROCEDURE

- ① Map each drug to the targets it acts on.
- ② Group drugs that share targets into clusters.
- ③ Give each child whole clusters — never split them.
- ④ Keep total drug count within the 4-drug limit.

Key finding: Removing MCX hurts performance more than removing any other component.

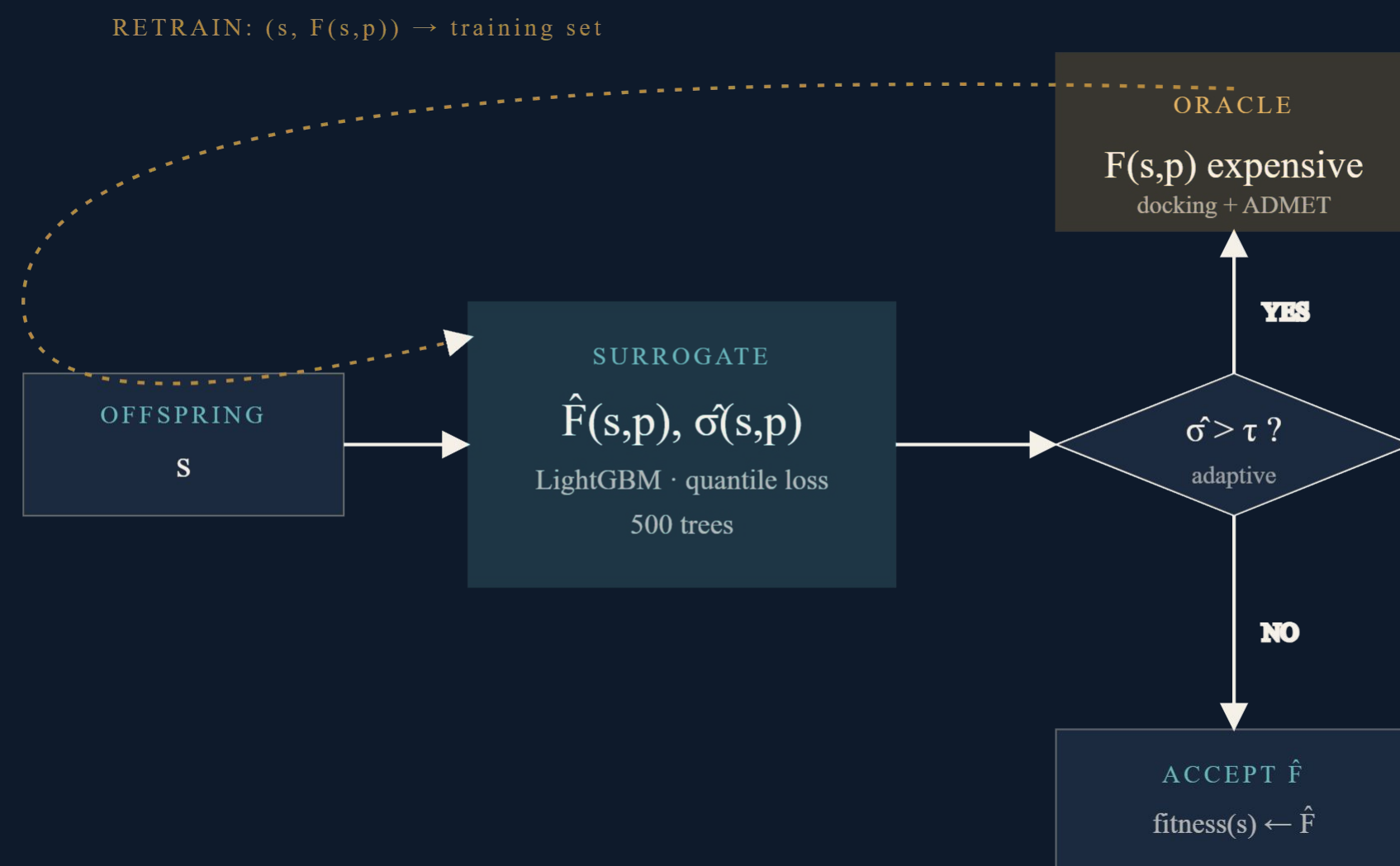
DRUG → TARGET BIPARTITE GRAPH



Cluster A → child 1 · Cluster B → child 2
Intact pharmacology preserved.

08 · SURROGATE

A fast shortcut — with a safety check.



38×

faster than running full simulations every time

94%

agreement with the full simulation — almost no quality lost

~2/3

of candidates scored by the fast model, not the slow one

Benchmark & experimental setup.

COMPOUNDS

I,842

DrugBank 5.1 + ChEMBL 33 ·
Lipinski-filtered

PATIENTS (SIMULATED)

500

1000 Genomes × Synthea
comorbidities

THERAPEUTIC AREAS

4

HTN · NSCLC · T2DM · MDD

ORACLE BUDGET · SEEDS

5k · 30

calls per run · averaged

ACTIONABLE PHARMACOGENES SAMPLED

CYP2D6

CYP2C19

CYP2C9

CYP3A5

SLC01B1

TPMT

DPYD

UGT1A1

VKORC1

HLA-B

HLA-A

NAT2

BASELINES (SAME BUDGET)

Random search

NSGA-II

SPEA2

MOEA/D

PG-RL · Transformer policy

10 · RESULTS

PharmaGA dominates on every metric.

METHOD	HV ↑	IGD ↓	PF ↑	HV / 1K ORACLE	RUNTIME (S)
Random	0.312	0.287	0.54	0.062	41
NSGA-II	0.601	0.141	0.63	0.120	188
SPEA2	0.589	0.148	0.61	0.118	204
MOEA/D	0.612	0.133	0.65	0.122	172
PG-RL	0.578	0.158	0.67	0.116	395
PharmaGA	0.688	0.097	0.82	0.244	167

Mean over 30 seeds × 500 patients × 4 areas. HV reference (1.1, 1.1, 1.1, 1.1) after per-objective normalization.

HYPERVOLUME

+12.4%

vs. MOEA/D (best baseline)

IGD

-27.1%

lower is better — reduction vs. MOEA/D

PERSONALIZATION FIDELITY

0.65 → 0.82

the largest gap — and the headline finding

11B • PARALLEL COMPUTING

Why parallel computing is needed for this.

BOTTLENECK 1 • POPULATION EVALUATION

Each round tests **400 combinations**. About 130 need slow simulations — each is independent, so they run in parallel.

BOTTLENECK 2 • PATIENT COHORT

Each patient gets their own independent search — **2,000 runs** with no communication between them.

BOTTLENECK 3 • SURROGATE RETRAINING

The prediction model retrains every round using multiple cores — already built in.

BOTTLENECK 4 • CLINICAL SCALE

Real hospitals have 10,000+ drugs and millions of patients. Parallel computing makes this scale practical.

PROJECTED WALL-CLOCK TIME AT B = 5,000 (SINGLE PATIENT RUN)

1 core (sequential) ~167 s

8 cores (oracle parallelism) ~32 s

Full cohort, 64 cores ~38 s / patient

"Parallel computing transforms PharmaGA from a theoretical model into a clinically viable system."

Where do the gains come from?

VARIANT	HV ↑	IGD ↓	PF ↑
Full PharmaGA	0.688	0.097	0.82
– Surrogate (oracle only)	0.652	0.111	0.81
– MCX (uniform crossover)	0.631	0.128	0.74
– Patient niching	0.672	0.102	0.69
– Pharmacogenomic conditioning	0.665	0.109	0.58
– Adaptive mutation	0.676	0.104	0.80

Oracle budget fixed at B = 5,000 for every variant.

DOMINANT DRIVER – HV / IGD

MCX.

Largest single drop when removed.

DOMINANT DRIVER – PF

Patient gene profiling + niching.

Personalization score drops to 0.58 without it.

WHAT PF EXPOSES

Standard scores can look fine while the algorithm ignores patient differences. Only PF catches this.

12 · CASE STUDY

Same age, same weight — but different genes. Different drugs.

PATIENT A

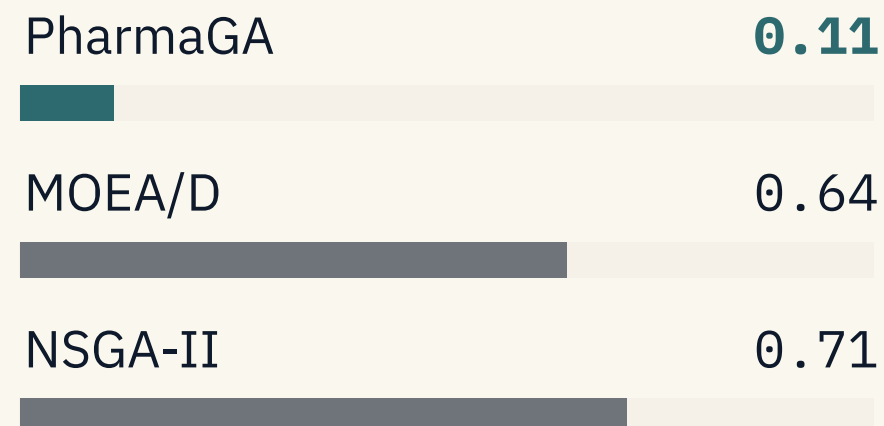
Age 62 · BMI 28 · eGFR 68

CYP2D6 · ULTRA-RAPID METABOLIZER

PHARMAGA RETURNS

Dose-escalated
venlafaxine-containing
regimens.

PARETO-FRONT JACCARD OVERLAP



PharmaGA gave the two patients almost completely different drug lists. Other methods gave them nearly the same one.

PATIENT B

Age 62 · BMI 28 · eGFR 68

CYP2D6 · POOR METABOLIZER

PHARMAGA RETURNS

CYP2D6-independent, **sertraline**
/ mirtazapine regimens.

13 • LIMITATIONS & DIRECTIONS

Honest limits — and where we go next.

LIMITATIONS

- L1** Simulated patient profiles
Tested on fake patients only. Real patient data still needed.
- L2** Incomplete drug–target graph
Drug databases are incomplete. Unknown interactions can't be caught.
- L3** Weakest surrogate — MDD
Depression predictions are the least reliable — hard to model from biology alone.

FUTURE WORK

- F1** Retrospective cohort validation
Test on real patient records and compare results to what doctors actually prescribed.
- F2** Temporal regimens
Extend to treatment phases over time, not just a one-time drug pick.
- F3** Conformal-calibrated surrogates
Make the fast model's confidence scores statistically trustworthy.

14 • CONCLUSION

PharmaGA picks better drugs — for the right patient, fast.

01

Patient genes are part of the goal. The algorithm optimizes for *this* patient from the start — not as an afterthought.

02

Drug combinations that make sense. MCX keeps drug pairs that work together — instead of mixing them randomly.

03

A fast model does the heavy lifting. Expensive simulations only run when truly needed — 38× faster overall.

This approach also works for drug repurposing — finding new uses for existing drugs, especially for rare diseases with little trial data.

THANK YOU — QUESTIONS WELCOME